

Sec. 2.2: Introduction to proofs

- A **theorem** is a statement that can be proven to be true.
- A **proof** consists of a series of steps, each of which follows logically from assumptions, or from previously proven statements, whose final step should result in the statement of the theorem being proven.
- An **axiom** is a statement assumed to be true.

Many mathematical theorems are universal statements or implication statements.

In this course, we will introduce the following methods of proof:

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| (a) Proof by exhaustion ($\forall$) | (e) Direct proof ($\rightarrow$, $\leftrightarrow$) |
| (b) Disproof by counterexample ($\forall$, $\rightarrow$) | (f) Prove by contrapositive ($\rightarrow$, $\leftrightarrow$) |
| (c) Constructive proof of existence ($\exists$) | (g) Proof by cases |
| (d) Proof by contradiction | (h) Proof by induction |

Recall:

	To prove	To disprove
$\forall x, P(x)$	Show that $P(x)$ is T for every x in the domain.	Find one counterexample x such that $P(x)$ is F .
$\exists x, P(x)$	Find one example x such that $P(x)$ is T .	Show that $P(x)$ is F for every x in the domain.

Proof by exhaustion: If the domain of a universal statement is small, it may be easiest to prove the statement by checking each element individually.

Example 1. Prove the statement: “For all positive integer $n < 3$, $(n + 1)^2 \geq 3^n$ ”

Disprove by Counterexample: To show that an implication or a “for all” statement is false, it is enough to find a single case (a counterexample) that the statement does not hold.

Example 2. Disprove each of the following statements.

- (a) If x is a real number and $x^2 < 1$, then $0 < x < 1$.
- (b) Every positive integer can be expressed as the sum of the squares of two integers.

Constructive proof of existence: To show that a “there exists” statement is true, we can give a specific example of an element in the domain or a set of directions to construct an element in the domain that has the required properties.

Example 3. Prove each of the following statements:

- (a) There are integers c and d such that $3c + 5d = 1$.
- (b) There is an integer n such that $2^n - 1$ is prime.

Prove by contradiction: To prove a statement $\mathcal{A}$ is true:

1. Assume that $\neg\mathcal{A}$ is true.
2. If this leads to a false statement (a contradiction), then $\neg\mathcal{A}$ is false. Thus, $\mathcal{A}$ must be true.

Example 4. Prove the statement: $\mathcal{A}$: “There is no largest integer. ”

1. Assume $\neg\mathcal{A}$: “There is a largest integer. ”
2. Denote this largest integer by N . Then $N \geq m$ for all integers m .
Consider $\boxed{}$, which is also an integer because it is the sum of two integers.
Then $\boxed{}$, which contradicts our assumption. Thus $\neg\mathcal{A}$ is false, so $\mathcal{A}$ is true.